

Combined Stylistic Clustering: Poets + Quran

Results Summary — Experiments 1 & 2

Objective

Two experiments extending the pre-Islamic poetry stylometry pipeline (Arabic-SBERT embeddings → UMAP → HDBSCAN clustering) to include the Quran, to see how it positions relative to a corpus of 260 known, distinct human poets.

Experiment 1: the entire Quran treated as one author (114 surahs = its 114 works) mixed with the 260 poets → 261 total entities.

Experiment 2: each of the 114 surahs treated as its own separate author, mixed with the 260 poets → 374 total entities.

Results — Experiment 1 (N = 261)

Metric	Value
Total entities	261 (260 poets + Quran)
Clusters found	13
Outliers	40
Silhouette score	0.469
Quran's cluster	Cluster 1 (19 poets share it)
Top similarity to nearest poet	0.976 (cosine)

Key finding: the Quran did NOT form its own isolated cluster. It landed inside Cluster 1 alongside 19 real poets, with cosine similarity to its five nearest poets ranging 0.971–0.976 — markedly higher than the poetry corpus's own average pairwise similarity (0.80, from the base paper). Most poets in that cluster carry the Fazari tribal name.

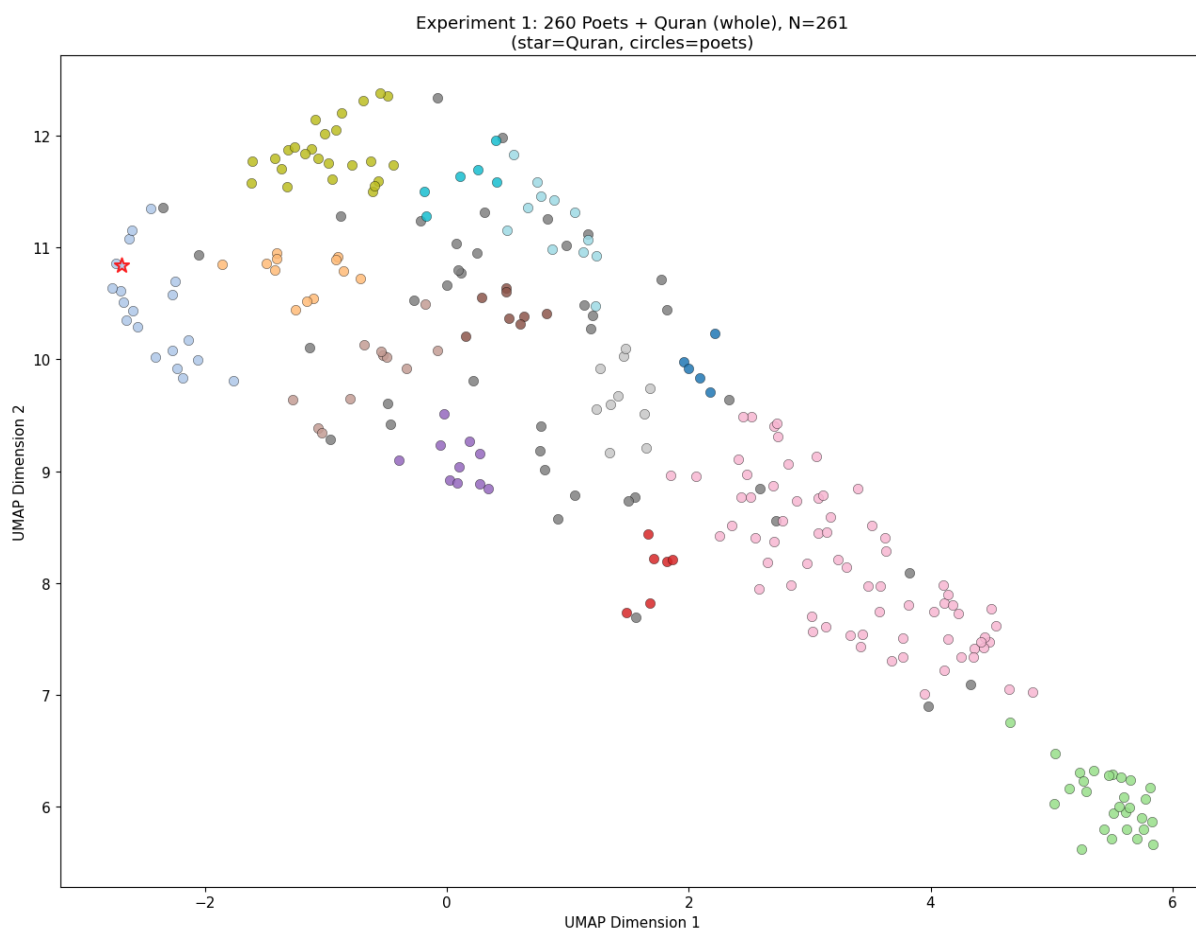


Figure 1. UMAP projection, Experiment 1. Red star = Quran.

Results — Experiment 2 (N = 374)

Metric	Value
Total entities	374 (260 poets + 114 surahs)
Clusters found	12
Outliers	11 (10 poets + 1 surah)
Silhouette score	0.469
Surah-only clusters (no poets)	3 clusters, 84 surahs (74%)
Mixed clusters (poets + surahs)	4 clusters, 30 surahs (26%) + 26 poets
Surahs classified as outliers	1 / 114

Key finding: most surahs (84 of 114, 74%) cluster only with other surahs — consistent with the Quran having its own distinct register, as expected. But a substantial minority — 30 surahs (26%) — landed in clusters mixed with a small, recurring group of specific poets rather than with other surahs. Only 1 surah was an outlier, meaning surahs almost always found *some* group to belong to. Notably, the poets appearing in these mixed clusters substantially overlap with the same poets identified in Experiment 1.

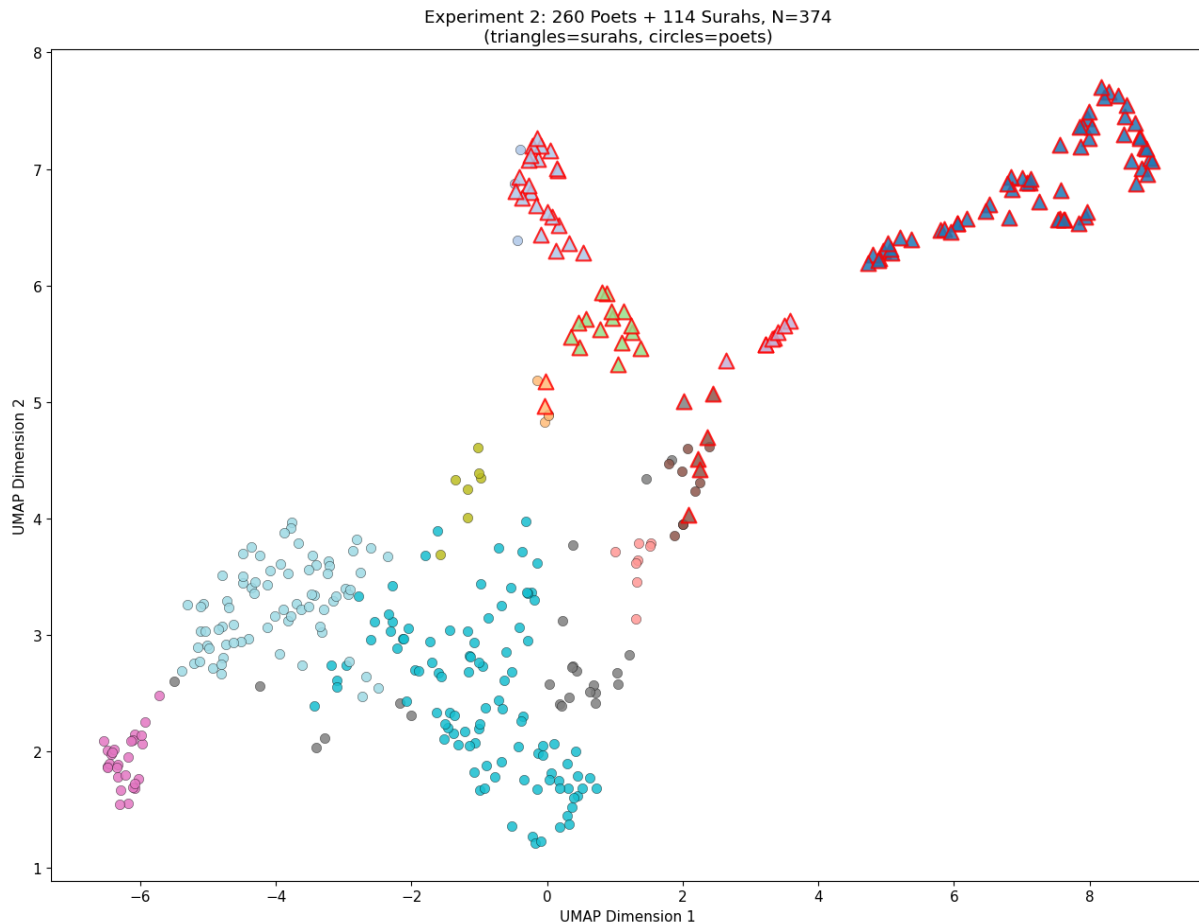


Figure 2. UMAP projection, Experiment 2. Red-outlined triangles = surahs.

Overall Verdict

The Quran does not read as maximally distinct from every single poet in this corpus. In both experiments, a consistent, small group of poets — largely from the same tribal naming pattern (Fazari) — sit closest to it in style-space, and roughly a quarter of individual surahs cluster with poets rather than exclusively with each other. At the same time, the majority of surahs (74%) do cluster together, separately from poets — so the dominant pattern is still "Quran as its own thing," with a notable minority exception.

One methodological note worth flagging

The poets who land closest to the Quran are, from the underlying poet data, generally minor figures with very few surviving poems. In the original poetry paper, poets with sparse output were shown to have no reliable relationship between output volume and stylistic centrality — but a small number of poems could still make a poet's averaged style vector less distinctive (closer to the corpus's overall center) than a well-represented poet's would be. That's a plausible partial explanation for why these particular poets keep appearing nearest to the Quran across both experiments — not necessarily that their actual writing style is unusually Quran-like. This wasn't tested directly here and would be worth checking (e.g. by re-running restricted to poets with a similar amount of surviving text as one surah).

What This Does and Does Not Prove

This measures writing-style similarity. It does not determine authorship. Stylistic overlap or distinctness between texts is consistent with many different explanations (shared genre conventions, shared language register, sample-size effects, or shared authorship) and this method cannot distinguish between them. The Quran landing near certain poets in embedding space is a description of measurable textual features, not evidence about who wrote what.